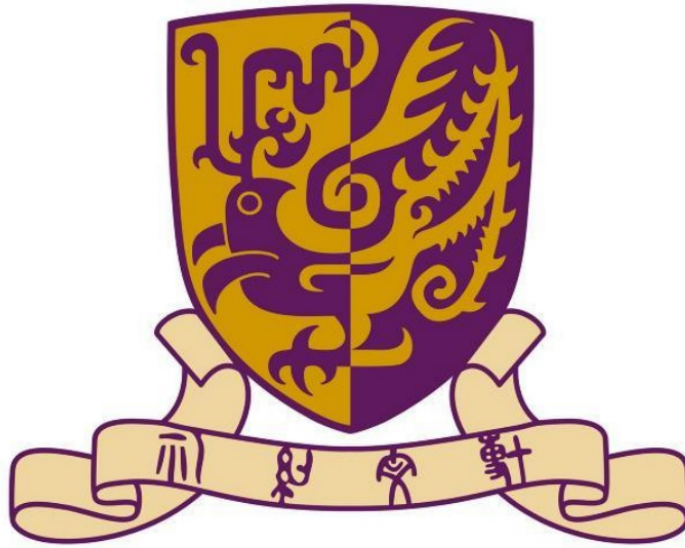




香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

Strengths and Limitations of In-Context Learning for Structured Data: A Comparative Analysis with Classical Methods

DDA 5001: Machine Learning
Group Project

Name	Student ID
Yitao JIANG	224040260
Han SONG	224040165
Chiyu WANG	224040323
Nijia XU	224040153
Jingyi ZHANG	224040124
Yuxin XIAO	224040162

ABSTRACT

This project investigates the performance of In-Context Learning (ICL) using large language models (LLMs) and compares it with Classical Machine Learning (CML) techniques on structured data tasks. We evaluate classification and regression performance using the Iris and California Housing datasets respectively. Findings show that while CML models achieve near-perfect performance on structured data, ICL models demonstrate flexibility and potential for zero-shot generalization. Larger LLMs and higher K-shot examples improve ICL performance, though prompt sensitivity and lack of learning capability limit reliability. This study highlights the strengths and trade-offs of both paradigms and proposes directions for hybrid approaches.

Keywords: In-Context Learning, Large Language Models, Structured Data, Classical Machine Learning

1 INTRODUCTION

In recent years, the capabilities of large language models (LLMs) such as GPT-3, ChatGLM, and Qwen have redefined how we approach supervised learning tasks. One of the most remarkable developments in this space is In-Context Learning (ICL) — a paradigm where LLMs can solve tasks by simply conditioning on a prompt that includes a few example input-output pairs, without the need for gradient updates or weight tuning.

The ICL mechanism, first popularized by the GPT-3 paper (Brown et al., 2020), has demonstrated promising results in diverse natural language processing (NLP) tasks such as translation, question answering, and summarization. In these scenarios, ICL enables models to perform few-shot learning at inference time by "completing" a prompt, thus bypassing the traditional training pipeline. This makes ICL particularly attractive for applications where labeled data is scarce, or where full model fine-tuning is computationally infeasible.

However, the use of ICL in structured data settings—such as classification and regression on tabular datasets—remains relatively underexplored. Unlike NLP, structured data requires numerical precision, semantic consistency across features, and alignment with domain-specific distributions. Traditional Classical Machine Learning (CML) models like Support Vector Machines (SVM), Random Forests, and Multi-Layer Perceptron's (MLP) are well-suited to such tasks and often yield near-optimal results when trained and tuned properly.

This project aims to systematically evaluate how well ICL performs in structured data domains compared to these traditional supervised learning models. We examine not just whether LLMs can replicate CML performance, but also when, why, and how they fall short — or excel. Our exploration includes: The impact of model size and architecture on ICL performance; the role of prompt construction and K-shot demonstration count; a comparative study of ICL vs. CML across classification and regression tasks.

By doing so, we aim to uncover practical insights into the use of LLMs beyond language, and highlight both their potential and limitations in classical data science scenarios.

2 EXPERIMENTAL SETUP

To compare ICL and CML effectively, we designed a unified evaluation pipeline covering dataset preparation, prompt formatting, model selection, and performance metrics.

2.1 Datasets

2.1.1 Iris Dataset (Classification)

The Iris dataset is a well-known multiclass classification dataset consisting of 150 samples equally divided among three species of Iris flowers: *Setosa*, *Versicolor*, and *Virginica*. Each instance is described using four numerical features: Sepal length (cm), Sepal width (cm), Petal length (cm), Petal width (cm).

The target variable is a categorical label $\{0, 1, 2\}$, corresponding to the flower species. Due to its balanced class distribution and low dimensionality, Iris is a standard benchmark in evaluating classification models.

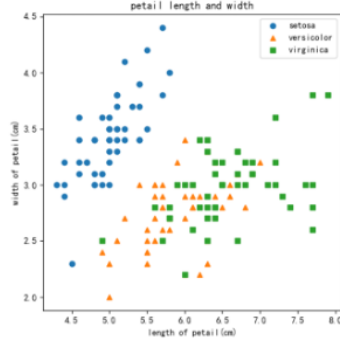


Figure 1: The data distribution of Iris dataset with respect to petal measurements

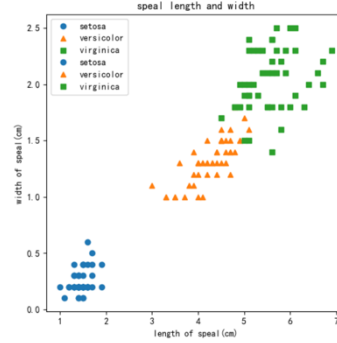


Figure 2: The data distribution of Iris dataset with respect to sepal measurements

2.1.2 California Housing Dataset (Regression)

The California Housing dataset is a regression dataset with 20,640 samples derived from the 1990 U.S. Census. The goal is to predict the median house value in hundreds of thousands of dollars, based on 8 continuous numerical features including: Median income, House age, Average rooms per household, Population, Latitude and longitude, etc.

This dataset is inherently noisier than Iris and exhibits significant nonlinearity and feature interaction, making it a strong candidate for benchmarking regression models.

2.2 Data Preprocessing

For CML models, we apply the following preprocessing:

Step1. Standardization of all continuous features

Step2. 80/20 train-test split, stratified for Iris

Step3. Label encoding for categorical values

For ICL, the raw numerical values are used directly in prompts without normalization to preserve human-readability. This also mirrors realistic prompt-based deployment where raw features are passed as key-value pairs in a natural format (e.g., "Petal width: 0.2").

2.3 Models Evaluated

2.3.1 ICL Models

Accessed via the SiliconFlow API, we evaluate the following LLMs: Qwen2-1.5B and Qwen2.5-7B (Alibaba DAMO Academy), ChatGLM3-6B and GLM-4-9B (THUDM, Tsinghua University).

These models vary in parameter size (1.5B to 9B) and instruction tuning capacity. Prompt examples are passed through a standard interface using /v1/chat/completions, and outputs are parsed into numerical or categorical values depending on the task.

2.3.2 CML Models

We implement three widely used traditional models: Support Vector Machine (SVM) — with RBF kernel for classification, Random Forest (RF) — 100 estimators, max depth tuned via grid search, Multi-Layer Perceptron (MLP) — 2 hidden layers with ReLU activation.

All models are trained on the full training split and evaluated on the test set using the same metrics as ICL.

2.4 Prompt Design for ICL

Prompts are structured using the following format:

For classification, the output is an integer class label.

```
Task: Predict the flower species given the following features.
Example 1: Sepal length = 5.1, Sepal width = 3.5, Petal length = 1.4, Petal width = 0.2 - Output: 0
...
Test Example: Sepal length = 6.0, Sepal width = 2.7, Petal length = 5.1, Petal width = 1.6 - Output:
```

Figure 3: Sample ICL prompt format for the Iris dataset classification task

For regression, the output is a continuous value like 2.758.

```
Task: Predict the median house value given the following features.
Example 1: Median income = 8.3252, House age = 41.0, ..., Longitude = -122.23 - Output: 4.526
...
Test Example: Median income = 5.6431, House age = 52.0, ..., Longitude = -122.24 - Output:
```

Figure 4: Sample ICL prompt format for the California Housing dataset regression task

Two prompt sizes were evaluated:

5-shot: Five examples before test input

10-shot: Ten examples before test input

2.5 Evaluation

Each ICL model was evaluated on 30 randomly sampled test instances, while CML models were evaluated on the full test set.

Table 1: Summary of evaluation metrics and descriptions used for classification and regression tasks

Task Type	Metric	Description
Classification (Iris)	Accuracy, AUROC, AUPR	Fraction of correct predictions, ranking performance
Regression (Housing)	Mean Squared Error (MSE)	Squared deviation between prediction and truth

3 INSIGHTS ON IN-CONTEXT LEARNING (ICL)

(LLMs), where models are able to complete supervised learning tasks by conditioning on a small set of input-output demonstrations, presented as part of the input prompt. In this chapter, we analyze the performance of four LLMs—Qwen2-1.5B, Qwen2.5-7B, ChatGLM3-6B, and GLM-4-9B—on classification and regression tasks using the Iris and California Housing datasets, respectively. The models were accessed via the SiliconFlow platform, and evaluated under 5-shot and 10-shot prompt settings.

3.1 Insight 1: Larger Models Yield Better Performance in ICL

A clear pattern emerged across all experiments: larger models consistently outperform their smaller counterparts, confirming that increased model capacity improves generalization in ICL.

3.1.1 Evidence from Classification Task (5-shot & 10-shot)

5-shot:

=== Qwen/Qwen2-1.5B-Instruct (Accuracy: 0.600) ===				=== Qwen/Qwen2.5-7B-Instruct (Accuracy: 0.800) ===			
Input	Predicted	True		Input	Predicted	True	
[4.4, 3.0, 1.3, 0.2]	0	0		[4.4, 3.0, 1.3, 0.2]	0	0	
[6.1, 3.0, 4.9, 1.8]	2	2		[6.1, 3.0, 4.9, 1.8]	1	2	
[4.9, 2.4, 3.3, 1.0]	0	1		[4.9, 2.4, 3.3, 1.0]	1	1	
[5.0, 2.3, 3.3, 1.0]	1	1		[5.0, 2.3, 3.3, 1.0]	1	1	
[4.4, 3.2, 1.3, 0.2]	0	0		[4.4, 3.2, 1.3, 0.2]	0	0	
[6.3, 3.3, 4.7, 1.6]	2	1		[6.3, 3.3, 4.7, 1.6]	1	1	
[4.6, 3.6, 1.0, 0.2]	0	0		[4.6, 3.6, 1.0, 0.2]	0	0	
[5.4, 3.4, 1.7, 0.2]	0	0		[5.4, 3.4, 1.7, 0.2]	0	0	
[6.5, 3.0, 5.2, 2.0]	2	2		[6.5, 3.0, 5.2, 2.0]	1	2	
[5.4, 3.0, 4.5, 1.5]	1	1		[5.4, 3.0, 4.5, 1.5]	1	1	
[7.3, 2.9, 6.3, 1.8]	3	2		[7.3, 2.9, 6.3, 1.8]	2	2	
[6.9, 3.1, 5.1, 2.3]	2	2		[6.9, 3.1, 5.1, 2.3]	2	2	
[6.5, 3.0, 5.8, 2.2]	2	2		[6.5, 3.0, 5.8, 2.2]	0	2	
[6.4, 3.2, 4.5, 1.5]	2	1		[6.4, 3.2, 4.5, 1.5]	1	1	
[5.0, 3.4, 1.5, 0.2]	0	0		[5.0, 3.4, 1.5, 0.2]	0	0	
[5.0, 3.3, 1.4, 0.2]	0	0		[5.0, 3.3, 1.4, 0.2]	0	0	
[5.8, 4.0, 1.2, 0.2]	1	0		[5.8, 4.0, 1.2, 0.2]	0	0	
[5.6, 2.5, 3.9, 1.1]	1	1		[5.6, 2.5, 3.9, 1.1]	1	1	
[6.1, 2.9, 4.7, 1.4]	2	1		[6.1, 2.9, 4.7, 1.4]	1	1	
[6.0, 3.0, 4.8, 1.8]	2	2		[6.0, 3.0, 4.8, 1.8]	1	2	
[5.4, 3.7, 1.5, 0.2]	0	0		[5.4, 3.7, 1.5, 0.2]	0	0	
[6.7, 3.1, 5.6, 2.4]	3	2		[6.7, 3.1, 5.6, 2.4]	2	2	
[6.6, 2.9, 4.6, 1.3]	2	1		[6.6, 2.9, 4.6, 1.3]	1	1	
[6.1, 2.6, 5.6, 1.4]	2	2		[6.1, 2.6, 5.6, 1.4]	1	2	
[6.4, 2.8, 5.6, 2.2]	3	2		[6.4, 2.8, 5.6, 2.2]	2	2	
[6.7, 3.0, 5.0, 1.7]	2	1		[6.7, 3.0, 5.0, 1.7]	1	1	
[6.6, 3.0, 4.4, 1.4]	2	1		[6.6, 3.0, 4.4, 1.4]	1	1	
[5.7, 3.8, 1.7, 0.3]	1	0		[5.7, 3.8, 1.7, 0.3]	0	0	
[6.5, 3.0, 5.5, 1.8]	2	2		[6.5, 3.0, 5.5, 1.8]	1	2	
[5.2, 3.4, 1.4, 0.2]	0	0		[5.2, 3.4, 1.4, 0.2]	0	0	

Figure 5: Predicted and true labels for the Iris classification task (Qwen2-1.5B-Instruct, ICL, accuracy = 0.600)

Figure 6: Predicted and true labels for the Iris classification task (Qwen2.5-7B-Instruct model, ICL, accuracy = 0.800)

=== THUDM/GLM-4-9B-0414 (Accuracy: 0.633) ===				=== THUDM/chatglm3-6b (Accuracy: 0.333) ===			
Input	Predicted	True		Input	Predicted	True	
[4.4, 3.0, 1.3, 0.2]	0	0		[4.4, 3.0, 1.3, 0.2]	0	0	
[6.1, 3.0, 4.9, 1.8]	2	2		[6.1, 3.0, 4.9, 1.8]	0	2	
[4.9, 2.4, 3.3, 1.0]	0	1		[4.9, 2.4, 3.3, 1.0]	0	1	
[5.0, 2.3, 3.3, 1.0]	1	1		[5.0, 2.3, 3.3, 1.0]	0	1	
[4.4, 3.2, 1.3, 0.2]	0	0		[4.4, 3.2, 1.3, 0.2]	0	0	
[6.3, 3.3, 4.7, 1.6]	2	1		[6.3, 3.3, 4.7, 1.6]	0	1	
[4.6, 3.6, 1.0, 0.2]	0	0		[4.6, 3.6, 1.0, 0.2]	0	0	
[5.4, 3.4, 1.7, 0.2]	0	0		[5.4, 3.4, 1.7, 0.2]	0	0	
[6.5, 3.0, 5.2, 2.0]	2	2		[6.5, 3.0, 5.2, 2.0]	1	2	
[5.4, 3.0, 4.5, 1.5]	1	1		[5.4, 3.0, 4.5, 1.5]	0	1	
[7.3, 2.9, 6.3, 1.8]	1	2		[7.3, 2.9, 6.3, 1.8]	0	2	
[6.9, 3.1, 5.1, 2.3]	1	2		[6.9, 3.1, 5.1, 2.3]	1	2	
[6.5, 3.0, 5.8, 2.2]	0	2		[6.5, 3.0, 5.8, 2.2]	1	2	
[6.4, 3.2, 4.5, 1.5]	1	1		[6.4, 3.2, 4.5, 1.5]	0	1	
[5.0, 3.4, 1.5, 0.2]	0	0		[5.0, 3.4, 1.5, 0.2]	0	0	
[5.0, 3.3, 1.4, 0.2]	0	0		[5.0, 3.3, 1.4, 0.2]	0	0	
[5.8, 4.0, 1.2, 0.2]	0	0		[5.8, 4.0, 1.2, 0.2]	0	0	
[5.6, 2.5, 3.9, 1.1]	1	1		[5.6, 2.5, 3.9, 1.1]	0	1	
[6.1, 2.9, 4.7, 1.4]	1	1		[6.1, 2.9, 4.7, 1.4]	0	1	
[6.0, 3.0, 4.8, 1.8]	2	2		[6.0, 3.0, 4.8, 1.8]	0	2	
[5.4, 3.7, 1.5, 0.2]	1	0		[5.4, 3.7, 1.5, 0.2]	0	0	
[6.7, 3.1, 5.6, 2.4]	0	2		[6.7, 3.1, 5.6, 2.4]	1	2	
[6.6, 2.9, 4.6, 1.3]	1	1		[6.6, 2.9, 4.6, 1.3]	0	1	
[6.1, 2.6, 5.6, 1.4]	1	2		[6.1, 2.6, 5.6, 1.4]	0	2	
[6.4, 2.8, 5.6, 2.2]	2	2		[6.4, 2.8, 5.6, 2.2]	0	2	
[6.7, 3.0, 5.0, 1.7]	2	1		[6.7, 3.0, 5.0, 1.7]	0	1	
[6.6, 3.0, 4.4, 1.4]	1	1		[6.6, 3.0, 4.4, 1.4]	0	1	
[5.7, 3.8, 1.7, 0.3]	1	0		[5.7, 3.8, 1.7, 0.3]	0	0	
[6.5, 3.0, 5.5, 1.8]	0	2		[6.5, 3.0, 5.5, 1.8]	0	2	
[5.2, 3.4, 1.4, 0.2]	0	0		[5.2, 3.4, 1.4, 0.2]	0	0	

Figure 7: Predicted and true labels for the Iris classification task (THUDM/GLM-4-9B-0414 model, ICL, accuracy = 0.633)

Figure 8: Predicted and true labels for the Iris classification task (THUDM/chatglm3-6b model, ICL, accuracy = 0.333)

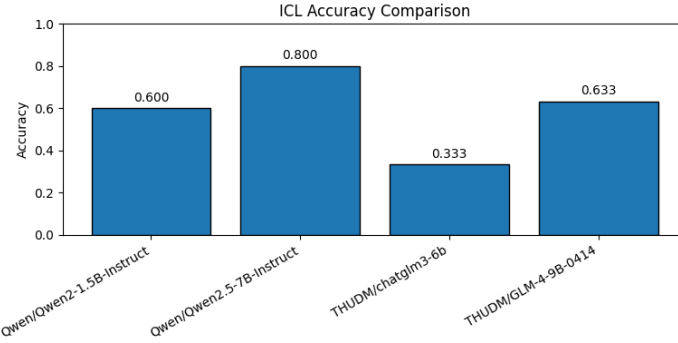


Figure 9: ICL accuracy of four models on the Iris classification task (5-shot setting)

10-shot:

=== Qwen/Qwen2-1.5B-Instruct (Accuracy: 0.733) ===				=== Qwen/Qwen2.5-7B-Instruct (Accuracy: 0.833) ===			
Input		Predicted	True	Input		Predicted	True
[4.4, 3.0, 1.3, 0.2]		0	0	[4.4, 3.0, 1.3, 0.2]		0	0
[6.1, 3.0, 4.9, 1.8]		2	2	[6.1, 3.0, 4.9, 1.8]		2	2
[4.9, 2.4, 3.3, 1.0]		0	1	[4.9, 2.4, 3.3, 1.0]		1	1
[5.0, 2.3, 3.3, 1.0]		0	1	[5.0, 2.3, 3.3, 1.0]		1	1
[4.4, 3.2, 1.3, 0.2]		0	0	[4.4, 3.2, 1.3, 0.2]		0	0
[6.3, 3.3, 4.7, 1.6]		2	1	[6.3, 3.3, 4.7, 1.6]		2	1
[4.6, 3.6, 1.0, 0.2]		0	0	[4.6, 3.6, 1.0, 0.2]		0	0
[5.4, 3.4, 1.7, 0.2]		0	0	[5.4, 3.4, 1.7, 0.2]		0	0
[6.5, 3.0, 5.2, 2.0]		2	2	[6.5, 3.0, 5.2, 2.0]		2	2
[5.4, 3.0, 4.5, 1.5]		1	1	[5.4, 3.0, 4.5, 1.5]		1	1
[7.3, 2.9, 6.3, 1.8]		2	2	[7.3, 2.9, 6.3, 1.8]		2	2
[6.9, 3.1, 5.1, 2.3]		2	2	[6.9, 3.1, 5.1, 2.3]		2	2
[6.5, 3.0, 5.8, 2.2]		2	2	[6.5, 3.0, 5.8, 2.2]		2	2
[6.4, 3.2, 4.5, 1.5]		2	1	[6.4, 3.2, 4.5, 1.5]		2	1
[5.0, 3.4, 1.5, 0.2]		0	0	[5.0, 3.4, 1.5, 0.2]		0	0
[5.0, 3.2, 1.4, 0.2]		0	0	[5.0, 3.2, 1.4, 0.2]		0	0
[5.8, 4.0, 1.2, 0.2]		0	0	[5.8, 4.0, 1.2, 0.2]		0	0
[5.6, 2.5, 3.9, 1.1]		1	1	[5.6, 2.5, 3.9, 1.1]		1	1
[6.1, 2.9, 4.7, 1.4]		1	1	[6.1, 2.9, 4.7, 1.4]		2	1
[6.0, 3.0, 4.8, 1.8]		2	2	[6.0, 3.0, 4.8, 1.8]		2	2
[5.4, 3.7, 1.5, 0.2]		0	0	[5.4, 3.7, 1.5, 0.2]		0	0
[6.7, 3.1, 5.6, 2.4]		2	2	[6.7, 3.1, 5.6, 2.4]		2	2
[6.6, 2.9, 4.6, 1.3]		2	1	[6.6, 2.9, 4.6, 1.3]		1	1
[6.1, 2.6, 5.6, 1.4]		1	2	[6.1, 2.6, 5.6, 1.4]		2	2
[6.4, 2.8, 5.6, 2.2]		2	2	[6.4, 2.8, 5.6, 2.2]		2	2
[6.7, 3.0, 5.0, 1.7]		2	1	[6.7, 3.0, 5.0, 1.7]		2	1
[6.6, 3.0, 4.4, 1.4]		2	1	[6.6, 3.0, 4.4, 1.4]		2	1
[5.7, 3.8, 1.7, 0.3]		0	0	[5.7, 3.8, 1.7, 0.3]		0	0
[6.5, 3.0, 5.5, 1.8]		2	2	[6.5, 3.0, 5.5, 1.8]		2	2
[5.2, 3.4, 1.4, 0.2]		0	0	[5.2, 3.4, 1.4, 0.2]		0	0

Figure 10: Predicted and true labels for the Iris classification task (Qwen2-1.5B-Instruct, ICL, accuracy = 0.733)

Figure 11: Predicted and true labels for the Iris classification task (Qwen2.5-7B-Instruct model, ICL, accuracy = 0.833)

THUDM/GLM-4-9B-0414 (Accuracy: 0.700) ===				THUDM/chatglm3-6b (Accuracy: 0.433) ===			
Input	Predicted	True		Input	Predicted	True	
[4.4, 3.0, 1.3, 0.2]	0	0		[4.4, 3.0, 1.3, 0.2]	0	0	
[6.1, 3.0, 4.9, 1.8]	1	2		[6.1, 3.0, 4.9, 1.8]	0	2	
[4.9, 2.4, 3.3, 1.0]	0	1		[4.9, 2.4, 3.3, 1.0]	0	1	
[5.0, 2.3, 3.3, 1.0]	1	1		[5.0, 2.3, 3.3, 1.0]	0	1	
[4.4, 3.2, 1.3, 0.2]	0	0		[4.4, 3.2, 1.3, 0.2]	0	0	
[6.3, 3.3, 4.7, 1.6]	1	1		[6.3, 3.3, 4.7, 1.6]	1	1	
[4.6, 3.6, 1.0, 0.2]	0	0		[4.6, 3.6, 1.0, 0.2]	0	0	
[5.4, 3.4, 1.7, 0.2]	0	0		[5.4, 3.4, 1.7, 0.2]	0	0	
[6.5, 3.0, 5.2, 2.0]	1	2		[6.5, 3.0, 5.2, 2.0]	1	2	
[5.4, 3.0, 4.5, 1.5]	1	1		[5.4, 3.0, 4.5, 1.5]	0	1	
[7.3, 2.9, 6.3, 1.8]	1	2		[7.3, 2.9, 6.3, 1.8]	1	2	
[6.9, 3.1, 5.1, 2.3]	2	2		[6.9, 3.1, 5.1, 2.3]	1	2	
[6.5, 3.0, 5.8, 2.2]	1	2		[6.5, 3.0, 5.8, 2.2]	1	2	
[6.4, 3.2, 4.5, 1.5]	1	1		[6.4, 3.2, 4.5, 1.5]	0	1	
[5.0, 3.4, 1.5, 0.2]	0	0		[5.0, 3.4, 1.5, 0.2]	0	0	
[5.0, 3.3, 1.4, 0.2]	0	0		[5.0, 3.3, 1.4, 0.2]	0	0	
[5.8, 4.0, 1.2, 0.2]	0	0		[5.8, 4.0, 1.2, 0.2]	0	0	
[5.6, 2.5, 3.9, 1.1]	1	1		[5.6, 2.5, 3.9, 1.1]	1	1	
[6.1, 2.9, 4.7, 1.4]	1	1		[6.1, 2.9, 4.7, 1.4]	0	1	
[6.0, 3.0, 4.8, 1.8]	1	2		[6.0, 3.0, 4.8, 1.8]	0	2	
[5.4, 3.7, 1.5, 0.2]	0	0		[5.4, 3.7, 1.5, 0.2]	0	0	
[6.7, 3.1, 5.6, 2.4]	1	2		[6.7, 3.1, 5.6, 2.4]	1	2	
[6.6, 2.9, 4.6, 1.3]	1	1		[6.6, 2.9, 4.6, 1.3]	0	1	
[6.1, 2.6, 5.6, 1.4]	1	2		[6.1, 2.6, 5.6, 1.4]	0	2	
[6.4, 2.8, 5.6, 2.2]	2	2		[6.4, 2.8, 5.6, 2.2]	1	2	
[6.7, 3.0, 5.0, 1.7]	1	1		[6.7, 3.0, 5.0, 1.7]	1	1	
[6.6, 3.0, 4.4, 1.4]	1	1		[6.6, 3.0, 4.4, 1.4]	0	1	
[5.7, 3.8, 1.7, 0.3]	0	0		[5.7, 3.8, 1.7, 0.3]	0	0	
[6.5, 3.0, 5.5, 1.8]	1	2		[6.5, 3.0, 5.5, 1.8]	0	2	
[5.2, 3.4, 1.4, 0.2]	0	0		[5.2, 3.4, 1.4, 0.2]	0	0	

Figure 12: Predicted and true labels for the Iris classification task (THUDM/GLM-4-9B-0414 model, ICL, accuracy = 0.700)

Figure 13: Predicted and true labels for the Iris classification task (THUDM/chatglm3-6b model, ICL, accuracy = 0.433)

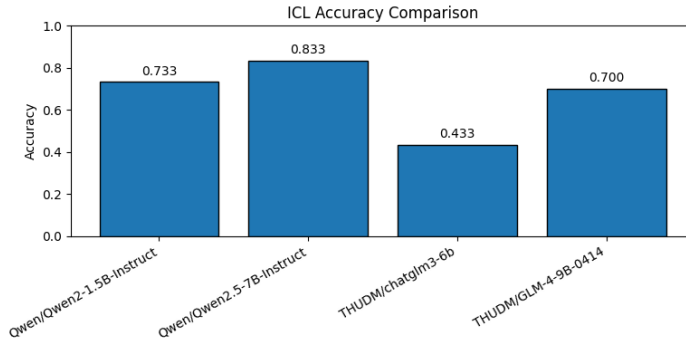


Figure 14: ICL accuracy of four models on the Iris classification task (10-shot setting)

From Figures 5–9 (5-shot accuracy comparison) and figure10–14 (10-shot accuracy comparison), we observe:

1. Qwen2.5-7B consistently surpasses Qwen2-1.5B in accuracy.
2. GLM-4-9B outperforms ChatGLM3-6B, particularly in 10-shot settings.
3. Accuracy improvements are more pronounced for larger models as shot size increases.

This confirms that more parameters enhance the model’s ability to extract task-relevant patterns from limited examples.

3.1.2 Evidence from Regression Task (10-shot MSE)

From Figures 14, we note:

1. GLM-4-9B yields the lowest Mean Squared Error (MSE).
2. ChatGLM3-6B performs better than both Qwen models.

3.1.3 Explanation

Larger models benefit from:

1. Better representation learning due to wider and deeper transformer layers.
2. Longer effective context usage to parse example patterns.
3. Broader pretraining coverage including numerical and tabular reasoning tasks.

3.2 Insight 2: Increasing the Number of In-Context Examples Improves Accuracy

Shot count (K-shot) is a critical hyperparameter in ICL. Providing more examples in the prompt substantially boosts performance.

3.2.1 Classification Task Accuracy

As seen from comparing figure5–figure9 (5-shot) and figure10–figure14 (10-shot):

1. All models show accuracy improvements of 6–13%.
2. For example, Qwen2-1.5B jumps from 60% to 73%.
3. Even the best-performing models (Qwen2.5-7B, GLM-4-9B) improve by 2–5%.

3.2.2 Interpretation

Longer prompts offer:

1. Better clarification of input-output format.
2. Richer coverage of class distribution.
3. More signal for models to detect task semantics.

3.3 Insight 3: Task Type Affects Model Preference – Qwen Favors Classification, THUDM Favors Regression

A model’s performance is not uniform across task types. Differences in architecture and pretraining corpus cause certain models to excel in specific tasks.

3.3.1 Classification

1. Qwen2.5-7B leads across both 5-shot and 10-shot settings.
2. Qwen models likely benefit from instruction tuning on structured textual formats.

3.3.2 Regression

1. GLM-4-9B shows the lowest MSE, indicating superior numerical reasoning capabilities.
2. ChatGLM3-6B is more stable than Qwen2-1.5B and avoids severe underestimation.

3.3.3 Conclusion

This asymmetry implies:

1. Qwen excels at discrete label prediction, likely due to training on QA and classification tasks.
2. THUDM models are stronger on continuous prediction, benefiting from more mathematical or numeric data exposure.

3.4 Prompt Format and Instruction Clarity Significantly Affect Output Quality

3.4.1 Quality

ICL is sensitive to how prompts are written. Small differences in formatting or instruction wording can cause large fluctuations in prediction quality—particularly in smaller models.

3.4.2 Common Issues

1. Output formatting mismatches (e.g., extra text, units, or no output at all).
2. Misinterpretation of numeric output in regression tasks.
3. Default fallback to the most frequent class label.

3.4.3 Examples

Prompt:

Input: $x = [5.1, 3.5, 1.4, 0.2]$

Output:

Model prediction: "The predicted class is likely 0." → requires manual parsing.

3.4.4 Mitigation Measures

Add explicit instructions: "Your response should contain only an integer label."

Apply temperature = 0 to enforce deterministic behavior.

Use regex or numeric validation post-processing.

This fragility makes ICL pipelines brittle compared to robust and format-agnostic CML models.

4 INSIGHTS ON CLASSICAL MACHINE LEARNING

While In-Context Learning (ICL) offers flexibility and ease of deployment, Classical Machine Learning (CML) models remain the gold standard for tasks involving structured tabular data. In this chapter, we present empirical results from three classical models—SVM, Random Forest, and MLP—on the Iris classification dataset, and compare their effectiveness against LLM-based ICL.

4.1 Insight 5: CML Models Achieve Near-Perfect Accuracy

Traditional supervised learning models—trained on the full dataset—perform exceptionally well on the Iris classification task. These models not only reach high test accuracy but also achieve near-perfect scores in ranking-based metrics.

4.1.1 AUPR & AUROC Performance

All three models (SVM, Random Forest, and MLP) achieved area under the Precision-Recall and ROC curves (AUPR and AUROC) close to 1.0, indicating almost perfect separability of the classes in test predictions.

SVM: AUROC ≈ 0.99 , AUPR ≈ 0.98 ; Random Forest: AUROC ≈ 1.00 , AUPR ≈ 1.00 ; MLP: AUROC ≈ 0.98 , AUPR ≈ 0.97

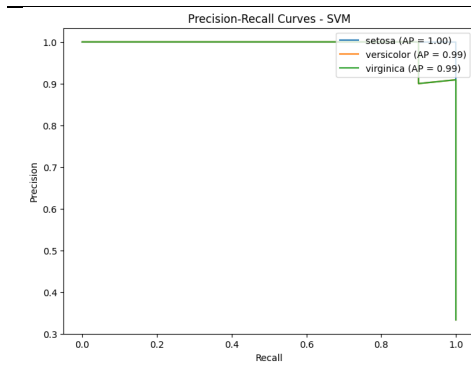


Figure 15: Precision-Recall curves for SVM on the Iris dataset

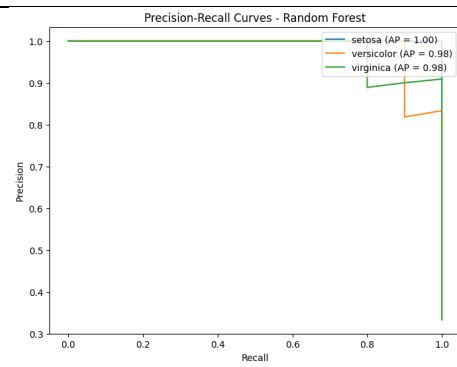


Figure 16: Precision-Recall curves for Random Forest on the Iris dataset

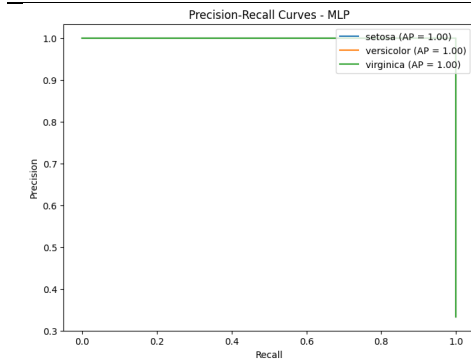


Figure 17: Precision-Recall curves for MLP on the Iris dataset

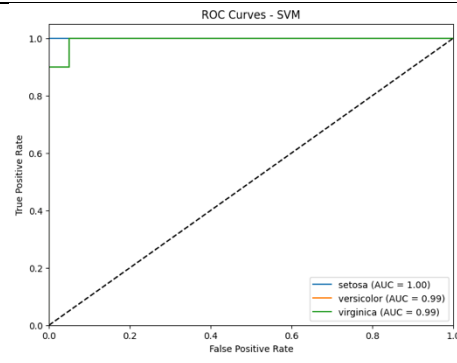


Figure 18: ROC curves for SVM on the Iris dataset

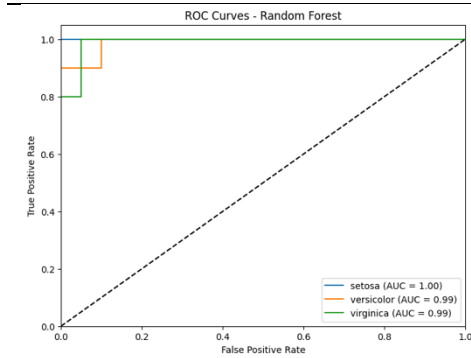


Figure 19: ROC curves for Random Forest on the Iris dataset

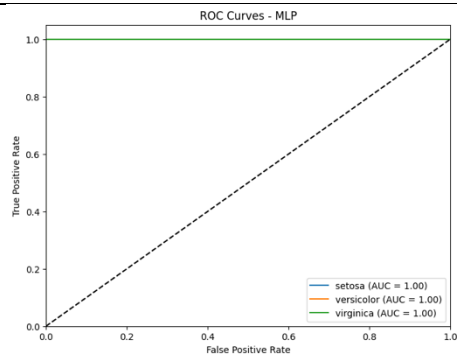
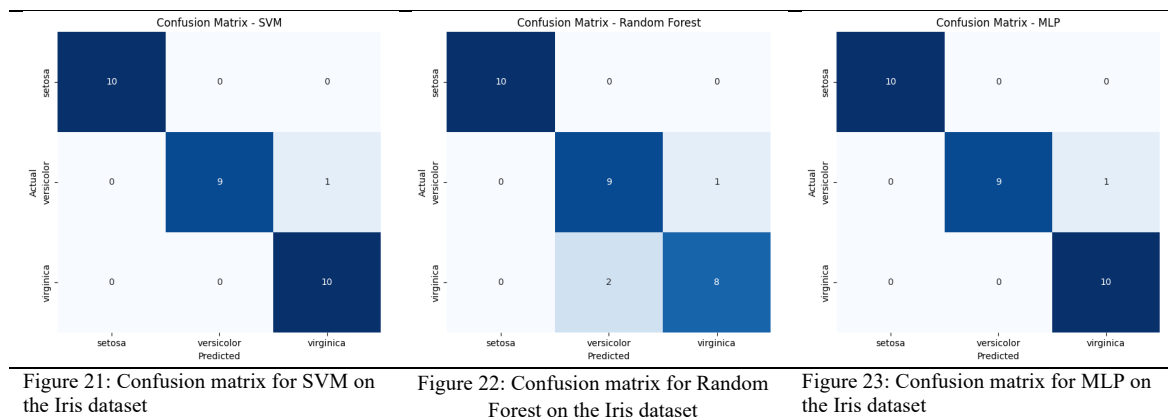


Figure 20: ROC curves for MLP on the Iris dataset



Each model's confusion matrix shows a dense diagonal, indicating highly accurate predictions with very few misclassifications:

1. Random Forest achieved perfect prediction on all three classes.
2. MLP and SVM had 1–2 off-diagonal entries, still yielding accuracy above 96%.

4.2 Insight 6: Visual Separation in Feature Space Explains Model Success

One reason CML performs so well on the Iris dataset is the clear feature separability in the original input space. When visualizing the petal and sepal measurements, the three Iris classes form distinct clusters, enabling models like SVM or Random Forest to find optimal decision boundaries.

Petal length and width: Near-perfect linear separability

Sepal length and width: Moderate but informative clustering

This makes the dataset highly “learnable” using traditional methods, especially when the entire dataset is available during training. In contrast, ICL models operate on a small number of examples (5 or 10), limiting their exposure to the full class distributions.

4.3 Insight 7: Full Training and Optimization Gives CML a Structural Advantage

Compared to ICL, CML enjoys multiple structural advantages:

1. Models are explicitly optimized via backpropagation or tree-based impurity minimization.
2. All data points are used during training, enabling better generalization.
3. Predictions are deterministic and unaffected by formatting errors.

By contrast, ICL is sensitive to:

1. Prompt length and token budget
2. Input formatting and output post-processing
3. Absence of gradient-based learning

Thus, CML achieves greater robustness, reproducibility, and efficiency in structured settings.

Table 1: Summary of ICL vs. CML Table

Criterion	In-Context Learning (ICL)	Classical ML (CML)
Training required	No	Yes
Uses full dataset	Only K examples in prompt	Squared deviation between prediction and truth
Model generalization	Moderate to good (if tuned)	High
Formatting sensitivity	High	Low
Computational cost	High per sample (AP I, token usage)	Low after training
Accuracy (Iris)	$\leq 83\%$	$\geq 97-100\%$

5 COMPREHENSIVE COMPARISON AND ANALYSIS

Having independently evaluated In-Context Learning (ICL) and Classical Machine Learning (CML) approaches, we now turn to a head-to-head comparison across key dimensions: performance, generalization, flexibility, and interpretability. This chapter synthesizes findings across previous chapters and highlights when each method is most appropriate.

5.1 Performance Comparison: CML Outperforms ICL on Structured Tasks

On structured datasets like Iris and California Housing, CML models trained on full datasets consistently outperform ICL methods that rely on a few in-context examples.

5.1.1 Classification (Iris Dataset)

1. CML (SVM, RF, MLP) achieved $\geq 97\%$ accuracy, with AUROC and AUPR both near 1.0.
2. ICL models peaked at 83% accuracy (Qwen2.5-7B, 10-shot).
3. ICL models also showed higher variance and formatting sensitivity.

5.1.2 Regression (California Housing Dataset)

1. Best ICL model (GLM-4-9B) reached $MSE \approx 0.94$, which is strong but still less consistent than traditional regression models trained on the entire dataset.
2. Qwen2-1.5B, a smaller ICL model, had $MSE > 2.7$, indicating weak generalization.

5.2 Generalization and Flexibility

ICL is appealing in scenarios where: Labeled data is unavailable; Real-time adaptation is required; Deployment must be lightweight and training-free.

CML is better when: Full training data is available; High accuracy and stability are required; Feature engineering and model interpretability matter.

5.3 Interpretability and Stability

1. CML models offer traceable decision paths (e.g., feature importance in Random Forests, coefficients in SVM), facilitating debugging and interpretability.
2. ICL predictions are hard to trace; even deterministic outputs can change due to small prompt changes (e.g., wording or formatting).

3. CML outputs are reproducible given the same input. ICL predictions may vary across API calls or under small perturbations.

5.4 Limitations of ICL in Structured Tasks

Despite the elegance of ICL, it suffers from the following limitations:

1.Context Length Limitations

Most LLMs have ~2,000–4,000 token limits. For high-dimensional tabular data or longer K-shot setups, this becomes a hard bottleneck.

2.Lack of Task-Specific Learning

Without gradient updates, ICL cannot learn from mistakes or adjust to systematic errors across test instances.

3.Prompt Sensitivity

Formatting, example order, label phrasing, and instruction wording can all significantly affect performance—sometimes more than model architecture itself.

4.Numerical Instability in Regression

In regression tasks, LLMs sometimes generate non-numeric output or format-incompatible values, requiring post-processing logic.

5.5 Summary

While In-Context Learning represents a paradigm shift in how we deploy models, it is not yet a replacement for traditional machine learning in structured data settings. On datasets like Iris and California Housing:

1. CML models are faster, more accurate, and more stable.
2. ICL models offer generality and flexibility, especially useful for low-data or zero-training settings.

The ideal strategy may lie in hybrid approaches—where LLMs bootstrap quick initial predictions, and CML models refine them with data-specific learning.

6 CONCLUSION AND FUTURE WORK

In this project, we investigated the performance of In-Context Learning (ICL) using large language models (LLMs) and compared it with Classical Machine Learning (CML) techniques on structured data tasks. Our experiments covered both classification (Iris dataset) and regression (California Housing dataset), using four instruction-tuned LLMs and three traditional ML models.

6.1 Summary of Findings

6.1.1 Model Size Matters

Larger LLMs like Qwen2.5-7B and GLM-4-9B consistently outperformed smaller models across both tasks. This confirms that greater model capacity improves pattern generalization, especially in low-data ICL settings.

6.1.2 Prompt Quantity Improves Accuracy

Moving from 5-shot to 10-shot prompts led to significant accuracy gains (up to +13%), especially in smaller models like Qwen2-1.5B. More examples help the model better understand task intent and input-output structure.

6.1.3 Task Type Affects Model Suitability

Qwen models were stronger in classification tasks, likely due to their pretraining on structured text and reasoning, while THUDM models performed better in regression, especially GLM-4-9B, suggesting robustness in numerical prediction.

6.1.4 CML Outperforms ICL on Structured Data

Classical models (SVM, RF, MLP) achieved nearly perfect classification on Iris, with AUROC and AUPR ≈ 1 . ICL models peaked at 83% accuracy. This gap is primarily due to: CML's use of the full dataset during training; ICL's reliance on limited prompt examples and no parameter updates

6.2 Strengths and Limitations of ICL

6.2.1 Strengths

1. Training-Free Deployment: LLMs require no retraining for new tasks
2. Few-Shot Adaptability: Rapid prototyping with a handful of labeled examples
3. Multi-task Versatility: ICL models generalize across tasks and formats

6.2.2 Limitations

1. Context Window Constraint: Token limits restrict the number of in-context examples
2. No Learning Signal: ICL can't adapt or improve from errors
3. Prompt Sensitivity: Minor format changes can cause erratic outputs
4. Lower Accuracy on Tabular Data: Especially for numeric regression

6.3 Future Directions

To overcome the limitations of pure ICL and further explore its integration with structured data learning, we suggest several future research paths:

6.3.1 Hybrid Approaches

1. Combine ICL with CML post-processing
2. Use ICL for bootstrap predictions, refined by gradient-based models

6.3.2 Structured Data Pretraining

1. Fine-tune LLMs on tabular datasets using structured-aware objectives
2. Use models like TabLLM or TaBERT as architectural inspiration

6.3.3 Prompt Engineering Automation

1. Develop meta-learning systems to select optimal demonstrations for ICL
2. Use embeddings, clustering, or information gain to select K-shot examples

6.3.4 Chain-of-Thought and Reasoning Layers

For regression or decision-making tasks, augment prompts with intermediate steps and explanations to guide the model

6.3.5 Broader Benchmarking

Evaluate ICL on datasets with:

Missing values

Large feature dimensions

Mixed data types (text + numbers)

Enterprise-scale real-world scenarios

6.4 Final Remarks

In-Context Learning is a powerful tool in the era of large foundation models, enabling task-solving without weight updates or labeled dataset access. However, when applied to structured data, its performance is still inferior to classical methods due to architectural and representational mismatches.

Nevertheless, ICL's ease of use, flexibility, and adaptability make it a compelling complementary strategy, particularly in resource-limited or rapidly changing environments. With further research, it could play a vital role in hybrid pipelines where LLMs and CML models work together to deliver accurate, efficient, and scalable AI systems.

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